Networks

Trees

Michel Bierlaire

Optimization: principles and algorithms

ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

Trees

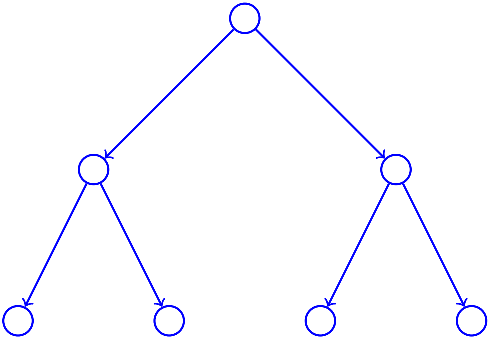
We introduce a family of graphs called "trees".
They are useful in many applications.

Trees

Definition

Tree

- Connected graph,
- without cycle.



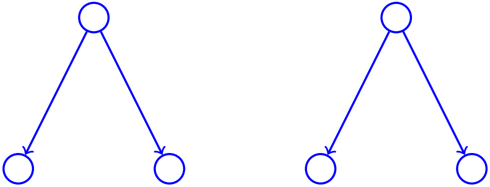
Forest

Mention that this is sometimes called a forest, as each connected component is a tree.

Definition

Not a tree

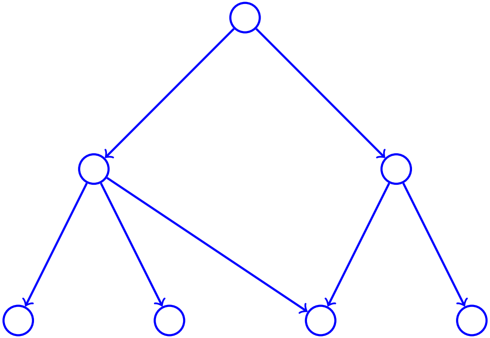
Not connected.



Definition

Not a tree

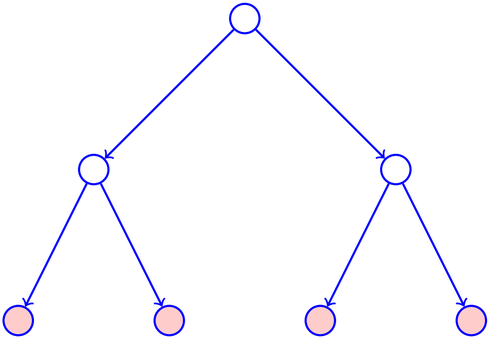
Contains a cycle.



Definition

Leaf

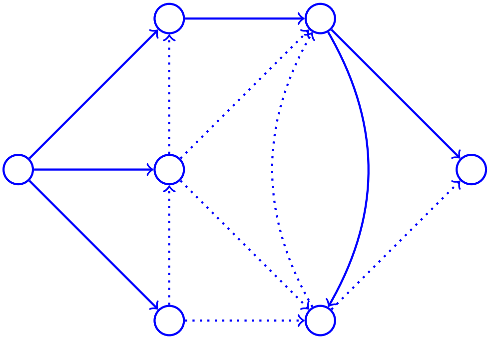
Node of degree 1.



Spanning tree

Definition

- Consider the graph $(\mathcal{V}, \mathcal{E}, \phi)$.
- The subgraph $(\mathcal{V}, \mathcal{E}', \phi')$
- is a spanning tree of $(\mathcal{V}, \mathcal{E}, \phi)$,
- if it is a tree.



Summary

Trees are connected graphs without cycle.
In the next video, we analyze some of their properties.